

Diagonalization 2

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Problem 1. *First, a sufficient condition. Suppose $P_A(\lambda)$ has $n := \dim V$ distinct roots, all in F . Show that the sum of the dim of the eigenspaces V_f over all f is at least n , and so conclude by the previous homework that its exactly n , and A is diagonalizable.*

Proof. Suppose $P_A(\lambda)$ has $n := \dim V$ distinct roots, all in F . By definition of the characteristic polynomial, there exists n distinct solutions to $\det(A - \lambda I) = 0$. By definition of invertibility, there exists $\lambda_1, \dots, \lambda_n \in F$ s.t $(A - \lambda_i I)$ is not invertible. Furthermore, it means $\ker(A - \lambda_i I) \neq \{0\}$. Suppose $v \in \ker(A - \lambda_1 I)$ and $v \in \ker(A - \lambda_2 I)$ s.t $v \neq 0$. Then it means $Av = \lambda_1 v = \lambda_2 v$, thus $\lambda_1 = \lambda_2$. Thus $V_{\lambda_i} = \ker(A - \lambda_i I)$ only share the 0 vector and have at least one other vector, by definition of eigenvalues. Moreover by the previous homework the vectors in distinct eigenspaces are linearly independent. Thus $\sum_{i=1}^n \dim V_{\lambda_i} \geq n$, because there are at least n distinct eigenspaces with a nonzero linearly independent vector. Thus $n \leq \sum_{i=1}^n \dim V_{\lambda_i} \leq \sum_f \dim V_f$. By the previous homework if the inequality is true we know that $n = \sum_f \dim V_f$. Thus A is diagonalizable. $\square$

Problem 2. *Give an example (a really simple one) of a diagonalizable matrix whose characteristic polynomial does not have distinct roots. Show that to be diagonalizable we at least need that $P_A(T)$ has all its roots in F , ie that $P_A(T)$ is a product of linear polynomials (factors completely), over F .*

Claim 2a. *There exists diagonalizable matrices whose characteristic polynomial doesn't have distinct roots.*

Solution. 0 is the example. It is by definition diagonalizable having scalars on the diagonal and zeros every where else. But its characteristic polynomial is $P_0(T) = T^n$ whose only roots are 0. $\blacksquare$

Claim 2b. *To be diagonalizable we at least need that $P_A(T)$ has all its roots in F , ie that $P_A(T)$ is a product of linear polynomials (factors completely), over F .*

Proof. Assume the contrary, $A : V \rightarrow V$ is a diagonalizable linear transformation whose characteristic polynomial $P_A(T)$ has roots which are not in the field F . By the previous homework, if A is diagonalizable there exists a eigenvector basis. For each eigenvector v , by definition, there exists a λ eigenvalue such that $Av = \lambda v$. By definition all the roots to $P_A(T)$ are eigenvalues. However since there exists a $c \notin F$ s.t $P_A(c) = 0$ thus c is a eigenvalue by definition. Thus there exists a v in our eigenvector basis such that $Av = cv$. However since $c \notin F$ then $Av = cv \notin V$. This forms a contradiction because the $A_*(V) \in V$ by definition of A . But there exists a $v \in V$ s.t $Av \notin V$. $A : V \rightarrow V$ is a diagonalizable linear transformation if its characteristic polynomial $P_A(T)$ only has roots in the field F . $\square$

Problem 3. Let the (distinct) eigenvalues of A be $f_1, \dots, f_k$. Consider the product $M := (A - f_1 I)(A - f_2 I) \dots (A - f_k I) = Q(A)$ where $Q(T) = (T - f_1) \dots (T - f_k)$. Show that the restriction of M to each eigenspace $V_{f_i}(A)$ is zero. Conclude that if A is diagonalizable, then $Q(A) = 0$.

Proof. For each eigenvalue f_i , let $V_{f_i}(A)$ be the corresponding eigenspace. By definition, the eigenspace $V_{f_i}(A)$ is the set of all eigenvectors associated with the eigenvalue f_i .

By definition, $V_{f_i}(A) = \ker(A - f_i I)$. $\forall v \in V_{f_i}(A)$ for all $i \in [1, \dots, k]$.

$$\begin{aligned} Mv &= (A - f_1 I)(A - f_2 I) \dots (A - f_k I)v \\ &= (A - f_1 I)(A - f_2 I) \dots (Av - f_k Iv) \\ &= (A - f_1 I)(A - f_2 I) \dots (f_i v - f_k v) \\ &= (A - f_1 I)(A - f_2 I) \dots (A - f_{k-1} I)(f_i - f_k)v \\ &= (f_i - f_k)(A - f_1 I)(A - f_2 I) \dots (A - f_{k-1} I)v \\ &= (f_i - f_k)(f_i - f_{k-1})(A - f_1 I)(A - f_2 I) \dots (A - f_{k-2} I)v \\ &\vdots \\ &= (f_i - f_1) \dots (f_i - f_k)(f_i - f_{k-1})v \end{aligned}$$

Since $v \in V_{f_i}(A)$ for $i \in [1, \dots, k]$. Then there exists a $j \in [1, \dots, k]$ where $i = j$ such that $f_i = f_j = 0$. Thus one factor of Mv is 0 thus $Mv = 0$

If A is diagonalizable, then there exists a basis of eigenvectors $B_v = \{v_1, \dots, v_{\dim V}\}$. Thus every $x \in V$ would be a linear combination of eigenvectors and since by properties of linear maps M upholds linearity, $Mx = \sum_{i=1}^{\dim V} a_i Mv_i$. However since v_i an element of some eigenspace, then $Mx = \sum_{i=1}^{\dim V} a_i 0 = 0$. Thus if A is diagonalizable, then for any $x \in V$, $Mx = 0$. Thus $Q(A) = 0$. $\square$

Problem 4. Conversely, suppose $f_1, \dots, f_k$ are distinct elements of F . For $Q(T)$ the poly with these roots, as in 3. Suppose $Q(A) = 0$. Use a lemma from an earlier homework (on the dim of the kernel of a product of matrices) to conclude that the sum of the dim of the Ker of the $A - f_i I$ is at least n . Show that we can drop any of the f_i that are not eigenvalues and everything still works. So we can assume all the f_i are eigenvalues. Show $\sum \dim V_{f_i} = n$. Conclude that A is diagonalizable, and that these are the only eigenvalues. Does $Q(T)$ have to be the characteristic polynomial?

Proof. Suppose $f_1, \dots, f_k$ are distinct elements of A . Let $Q(T)$ be a polynomial with these roots. Suppose $Q(A) = 0$. Since $Q(A) = 0$, $\ker(Q(A)) = V$ because for all $v \in V$, $Q(A)v = 0v = 0$. By Linear Algebra Bootcamp 4, we know that the dimensions kernel of the product of two matrices is less than the sum of dimensions of the kernel of each matrix. Thus $\dim(\ker Q(A)) = \dim(V) = n \leq \sum_{i=1}^k (\dim \ker(A - f_i I))$. By definition, $V_{f_i}(A) = \ker(A - f_i I)$. Thus $n \leq \sum_{i=1}^k (\dim V_{f_i}(A))$. If f_i for some $i \in [1, \dots, k]$ is not a eigenvalue, by definition of eigenvalues then $A - f_i I$ is invertible thus the kernel is the trivial kernel and the dimension of the kernel is 0. Thus $\dim V_{f_i}(A) = 0$ if f_i is not a eigenvalue and thus we can drop any $\dim V_{f_i}(A)$ from our sum if f_i is not a eigenvalue. Thus we can assume all f_i are eigenvalues wlog.

Since eigenvectors corresponding to distinct eigenvalues are linearly independent, the vectors in these distinct eigenspaces are all linearly independent from each other. Thus form a set vectors $B = \bigcup_{f_i \in [f_1, \dots, f_k]} B_{f_i}$ where B_{f_i} is the basis of $V_{f_i}(A)$. By definition of basis, the vectors in B_{f_i} are linearly independent and since vectors in distinct eigenspaces are linearly independent from each other then B

is a set of $\sum_{i=1}^k (\dim V_{f_i}(A))$ linearly independent vectors, because the dimension gives you the number of basis vectors in a space. Thus $\dim \text{span} B = \sum_{i=1}^k (\dim V_{f_i}(A))$. By Linear Algebra Bootcamp 1 problem 7, we have shown that a dimension of a subspace must be less than or equal to its parent space. Thus $\dim \text{span} B = \sum_{i=1}^k (\dim V_{f_i}(A)) \leq n$. If $n \leq \sum_{i=1}^k (\dim V_{f_i}(A)) \leq n$, then $\sum_{i=1}^k (\dim V_{f_i}(A)) = n$. Thus by Diagonalization 1 problem 7, $f_1, \dots, f_k$ are the only eigenvalues and A is diagonalizable. $\square$

Solution. No $Q(T)$ doesn't have to be the characteristic, it can be any polynomial s.t $Q(A) = 0$ and its roots are eigenvalues of A . $\blacksquare$

Claim 4c. A is diagonalizable iff there is a polynomial $Q(T)$ in $F(T)$, with distinct roots which splits completely over F (ie is a product of linear terms) such that $Q(A) = 0$

Proof. ($\implies$) Suppose A is diagonalizable. Then by Diagonalization 1 problem 4, since A is diagonalizable, there exists a eigenvector basis $E = w_1, \dots, w_n$ for V . By definition, since there exists eigenvectors, each eigenvector must also have a corresponding eigenvalue. Note the eigenvalue need not be distinct. Let $L = \{\lambda_1, \dots, \lambda_a\}$ be the unique eigenvalues of A where all $l \in L$ $l \in F$. Note because the eigenvalues do not need to be distinct then there may be eigenvectors in our basis which share eigenvalues in L . Then define, $Q(x) = \prod_{i=1}^a (x - \lambda_i)$ if x is a matrix then $Q(X) = \prod_{i=1}^a (X - \lambda_i I)$. By definition of Q , its roots are the unique eigenvectors of A thus Q has distinct roots which are in F . By definition of a basis, the eigenvector basis spans V thus $\forall v \in V$, $v = \sum_{i=1}^n c_i w_i$ where for every w_i there exists a $\lambda_{w_i} \in L$ s.t $w_i \in V_{\lambda}(A)$ and $c_i \in F$. Thus for all $v \in V$,

$$\begin{aligned} Q(A)v &= \prod_{j=1}^a (A - \lambda_j I) \left(\sum_{i=1}^n c_i w_i \right) \\ &= \sum_{i=1}^n \prod_{j=1}^a (A - \lambda_j I) (c_i w_i) \\ &= \sum_{i=1}^n c_i \prod_{j=1}^a (\lambda_{w_i} - \lambda_j) \\ &= 0 \end{aligned}$$

Since for every w_i there exists a $\lambda_{w_i} \in L$ s.t $w_i \in V_{\lambda}(A)$ and $c_i \in F$, for all $v \in V$, $Q(A)v = 0$. Thus $Q(A) = 0$.

($\impliedby$): Suppose there was a polynomial $Q(T)$ s.t with distinct roots which splits completely over F (ie is a product of linear terms) such that $Q(A) = 0$. Let the roots of Q be $x_1, \dots, x_a$. Since $Q(T)$ has distinct roots which split completely over F , it can be rewritten as $Q(T) = \prod_{i=1}^a (T - x_i)$. Furthermore since $Q(A) = 0$, then $\forall v \in V$ $Q(A)v = (\prod_{i=1}^a (A - x_i I))v = 0$. Then at least 1 factor $(A - x_i I)v = Av - x_i Iv = Av - x_i v = 0$. Consider each factor $(A - x_i I)$ seperately. If only 1 factor is 0, then v is eigenvector. Find all linearly independent vectors v for x_i s.t $(A - x_i I)v = 0$. These v would be the basis vectors of $V_{x_i}(A)$. Combine all the basis for all $V_{x_i}(A)$ for all $i \in [1, \dots, a]$. Then we a eigenvector basis thus by Diagonalization 1 problem 4 A is diagonalizable. $\square$

Problem 5. Let $F = \mathbb{C}$, the complex numbers. Suppose A is a square matrix of finite order, ie $A^k = I$ for some k . Show that A is diagonalizable, and moreover, that its eigenvalues are all roots of unity.

Proof. Let $F = \mathbb{C}$, the complex numbers. Suppose A is a square matrix of finite order, ie $A^k = I$ for some k . Let $Q(T) \in F(T)$, $Q(T) := T^k - 1$. By the Fundamental Theorem of Algebra, any complex polynomial has all solutions in the $\mathbb{C}$. Then $Q(A) = A^k - I = I - I = 0$.

Let λ be a eigenvalue of A . Then by definition there exists a $v \in V$, s.t $Av = \lambda v$. $A^{k-1}(Av) = A^{k-1}(\lambda v) = \lambda^k v$. $A^{k-1}(Av) = A^k v = Iv = v$. Thus $v = \lambda^k v$. Thus $\lambda^k = 1$. Thus λ are the roots of unity and there are k distinct λ thus there are k distinct eigenvalues. Furthermore each eigenvalue, is a root to $Q(T)$ because $Q(T) \in \mathbb{C}(T)$.

Thus by previous Theorem in problem 4, since Q has k distinct roots, A must be diagonalizable. $\square$

Problem 6. Suppose $Q(A) = 0$ for some polynomial $Q(T)$ in $F(T)$. Show that $Q(\lambda) = 0$ for every eigenvalue of A . In short, the eigenvalues satisfy any polynomial that A satisfies

Proof. Suppose λ is any eigenvalue of A . Then $\exists v \in V$ s.t $v \neq 0$, s.t $Av = \lambda v$. Then for all $P(T) \in F(T)$ where b is the degree of P , $P(A)v = (\sum_{i=0}^b c_i A^i)v = \sum_{i=0}^b c_i A^i v = \sum_{i=0}^b c_i \lambda^i v = P(\lambda)v$. Thus $P(A)v = P(\lambda)v$ for all $P(T) \in F(T)$. Let $Q(T) \in F(T)$ s.t $Q(A) = 0$. As previously shown $Q(A)v = Q(\lambda)v$. Thus $Q(\lambda)v = 0$ and $v \neq 0$. Thus $Q(\lambda) = 0$. Thus all eigenvalues of A are roots of Q . $\square$